Problems

De Broglie

1- calculate de Broglie's wavelength for:

a) an electron accelerated from rest through potential difference of 10^4 volts.

b) a grain of sand of mass 1 mg moving at ;i speed of 50 m/s.

2. calculate de Broglie's wavelength for:

a) 0.14 kg baseball moving with speed of 44 m/s.

b) a proton moving at the same speed and, '

c) an electron moving at the same speed.

3. Calculate the Broglie wavelength of an electron, proton and uranium atom, all having the same kinetic energy of 100 eV, $m_p = 1.067 \times 10^{-27}$ kg, and $m_u = 3.95 \times 10^{-25}$kg.

4. Calculate the kinetic energy of an electron to be confined to a nucleus and its de Broglie$^{i\#}$s wavelength would have to be less than $10^{'14}$ m.

According to the obtained result would one expect to find an electron in a nucleus?

Given that the electric potential energy of an electron distant by 10^{-1} m from a proton is U= -0.144 MeV

Heisenberg Uncertainty Principle

5. The position and momentum of 1 keV electron are simultaneously determined. If its position is located to within 1 A°, what is the percentage of the uncertainty in its momentum?

6. Obtain the uncertainty in the velocity of an electron confined within a cube of a volume: a)$10^{-9}m^3$ b)$10^{-l8}m^3$ c) $10^{-24}m^3$

7. An electron with kinetic energy E =4 eV is confined within a region whose linear

8.The time taken for an electron to drop fron an excited states to the ground state is about 10^{-8}s. If the energy difference is 1 eV find the uncertainty in(a) the energy of the emitted radiationthe(b) frequency of the radiation (c)the wavelength of the radiation Particle in an infinite a Particle in Box

9.Calculate the lowest possible energy of,

(a) An electron confined to moving in one-dimensional box of width 0.4nm,

(b) A proton confined to the same box.

(c) Write your comment about the obtained results.

10. A 7-eV electron is incident on a barrier 0 180 nm thick and 12 eV high. What is the probability that electron (a) will be tunnel through the barrier (b) will be reflected?

11. A laser emits light of wavelength equal 0.693.9 nm. If this light is due to transitions of an electron in a box from the n= 2 to the n=l state, find the width of the box.

12.Using the particle be in box model, calculate the lowest energy and momentum of an alpha particle(m_u 1.66x10 Kg) moving in a nucleus of diameter 4xl0-^{14}m.

13 An electron is confined to a 1 micron thin layer of silicon. Assuming that the semiconductor can be adequately descr bed by a one - dimensional quantum well with infinite walls. Calculate the lowest possible energy within the material in eV. If the energy is interpreted the kinetic energy of the electron, what is the corresponding electron velocity?